

Figures File for Assignment 3

March 27, 2026

Please refer to the attached pptx for animations.

Figures

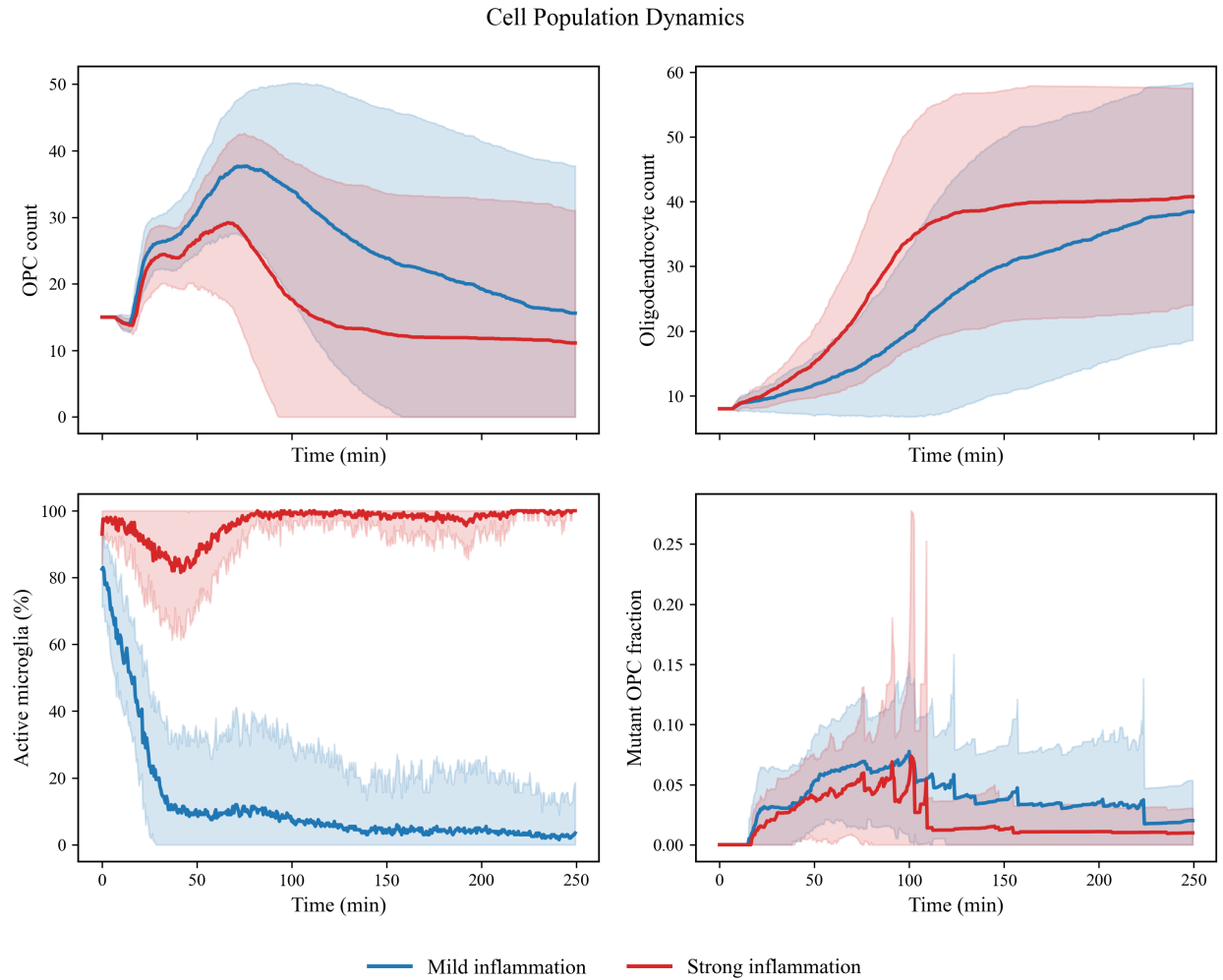


Figure 1: The plots show the temporal evolution of OPC counts, oligodendrocyte counts, microglia activation, and mutant OPC fraction under mild and strong inflammation regimes. Curves represent the ensemble mean across 25 simulation runs and shaded regions indicate the standard deviation.

Lesion Repair Dynamics

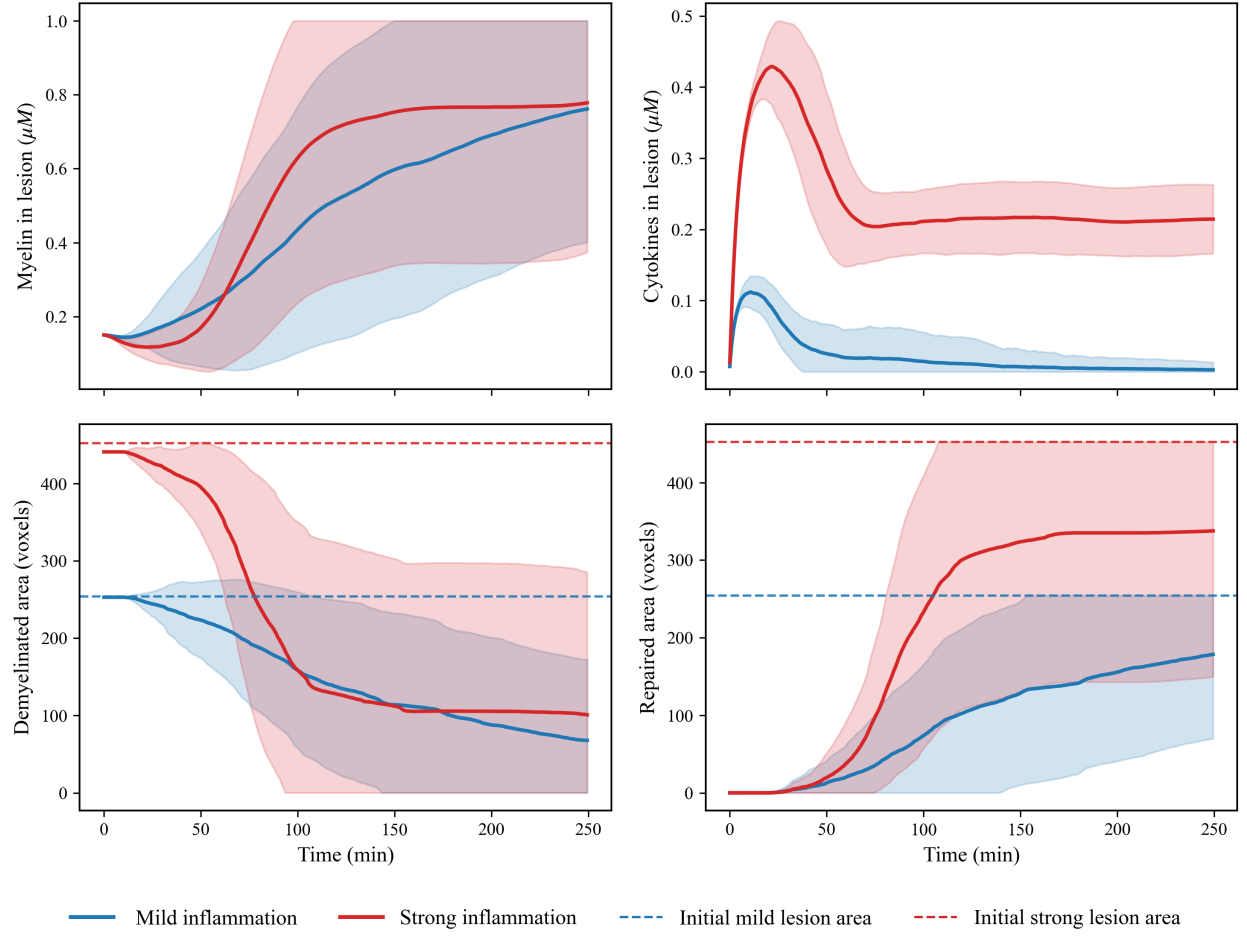


Figure 2: The panels show the temporal evolution of myelin levels, inflammatory cytokines, demyelinated area, and repaired area within the lesion region. Ensemble means across 25 simulations are shown with standard deviation shading. Dashed horizontal lines indicate the initial lesion area for the mild and strong regimes.

Local vs Global Biochemical Dynamics

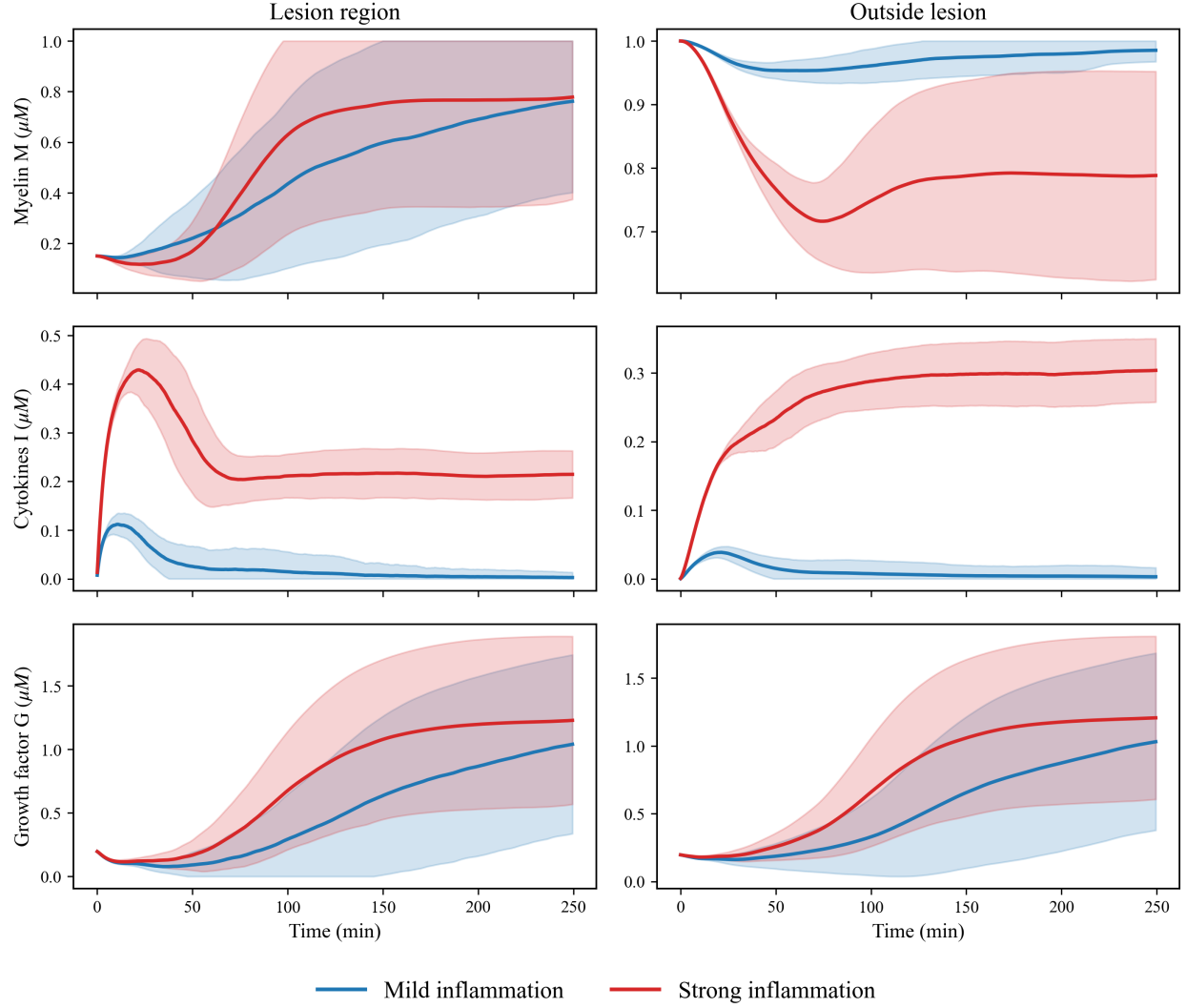


Figure 3: The left column shows the temporal evolution of myelin, cytokines, and growth factor concentrations within the lesion region, while the right column shows the same variables in the surrounding tissue. Curves represent ensemble means of 25 seeds with standard deviation shading.

Gene Regulatory Network Dynamics

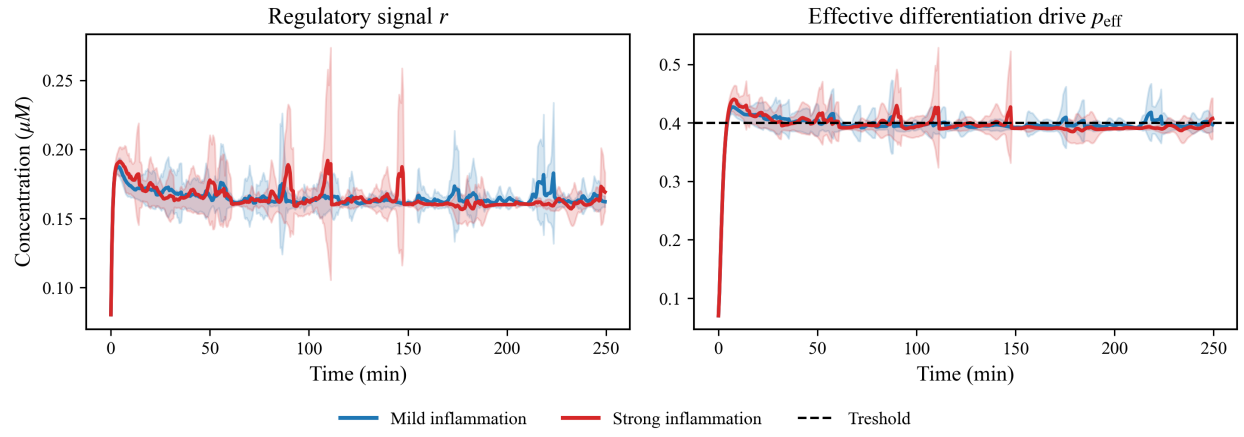


Figure 4: The plots show the evolution of the regulatory variable r , and the effective differentiation signal p_{eff} across 25 simulation runs. The dashed line indicates the differentiation threshold. Standard deviation bands are included.